Brian Xie

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EDUCATION

University of Maryland, College Park **Graduation:** May 2026

- B.S. Computer Science: Machine Learning **GPA:** 3.87/4.00
- B.S. Applied Mathematics

Honors & Distinctions: Banneker Key Full Ride Scholarship, ACES Honors Program

Coursework: Machine Learning | Computer Vision | Cryptography | Computational Methods | Linear Optimization | Theory & Methods of Statistics | Data Structures and Algorithms | Application of Linear Algebra | Advanced Calculus

EXPERIENCE

Technology Director | *Techncia, College Park* **January 2025 - November 2025**

- Led a team of **8 engineers** for the largest hackathon for **underrepresented genders**, supporting **1,500+ hackers**.
- Built and deployed RESTful web applications using **Nuxt.js**, AWS **DynamoDB**, **S3**, **Amplify**, and **Route 53** to manage event registration.
- Streamlined backend deployment with **AWS Serverless**, improving scalability and reducing operational overhead, while mentoring team members in Git workflows, code reviews, and cloud architecture best practices.

Software Engineer Intern | *Publicis Sapient, Illinois* **June 2025 - August 2025**

- Developed a financial chatbot integrated into a client website using **LangChain** (RAG), Firebase, Azure, and OpenAI, achieving a 2-second response time, supporting **chat preservation** for **context-aware responses**, and architected for future scalability.
- Refined prompts with licensed professional feedback to ensure **accuracy and compliance**, conducted unit and user testing, and collaborated in daily **Agile** standups to deliver secure, production-ready workflows.

Data Analysis Intern | *Earth System Science Interdisciplinary Center, College Park* **January 2025 - June 2025**

- Developed GUI-based satellite data analysis tools using **PyQt** and **Python**, automating ground observation retrieval and **cross-referencing datasets for validation**, enabling efficient multi-researcher usage.
- Participated in weekly standups, **pivoted development** for new features, and optimized data processing pipelines to improve accuracy and reduce manual effort on large-scale satellite datasets.

Teaching and Grading Assistant | University of Maryland **August 2025 - December 2025**

- Developed and evaluated **instructional problem solutions** for **35 students** in **Applications of Linear Algebra**, delivering detailed grading feedback and clarifying mathematical concepts.

PROJECTS

Java Fundamentals Debug Learning Application **April 2024 – April 2024**

- Developed a single-player educational game with **Java** and **Slick2D**, implementing **physics** simulation, diverse bug generation and **movement algorithms**, and **interactive UI** elements.
- Engineered gameplay mechanics that required strategic use of debugging tools to solve challenges, applying programming concepts in a practical, real-world context, and emphasizing problem-solving under complex constraints.

Linux Threat Actor Information Collection System **August 2023 – December 2023**

- Deployed **self-recycling** honeypots mimicking University systems, capturing **22,000 breach attempts** over **4 weeks** and enforcing permanent bans using NAT rules.
- Automated data collection and parsing with custom **Bash** and **Python scripts**, enabling analysis of attack patterns and identification of botnet leverage and common attack commands.

SKILLS

Languages: Python, Java, JavaScript, C, Bash, SQL, R, Racket

Framework and Tools: Vue.js, Nuxt.js, LangChain, AWS, Azure, Firebase, Git, Linux, Rest APIs